

How to Select Statistics for Research

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What determines statistical analysis

- Research questions
- Variables (level of measurement)
- Role of variables (independent, dependent variables and extraneous variables)
- Study design
- The number of independent and dependent variables
- Statistical assumption

Statistics vs Parameters

Parameters

- Property of population
- Example: height of Thai people, proportion of T2DM in Thai population

Statistics

- Property of sample
- Example: height of 100 randomly selected Thai people, proportion of T2DM from 400 people.

INFERENCE: → Parameters

SAMPLE DOMAIN

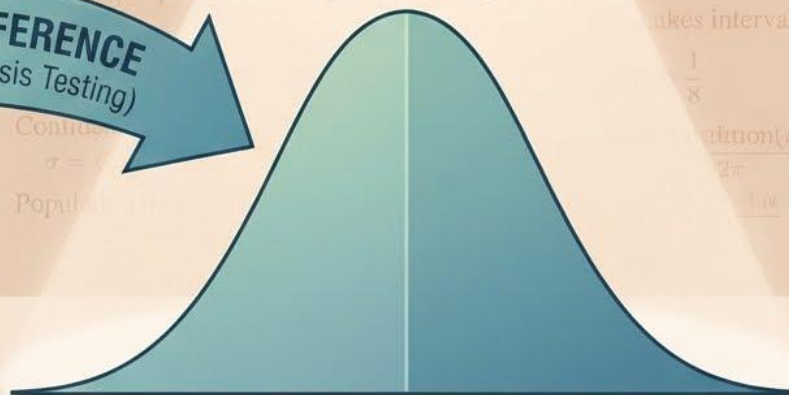


SAMPLE MEAN ($\bar{x}$) = 170.3 cm
SAMPLE STANDARD DEVIATION
(s) = 6.4 cm

STATISTICAL INFERENCE (Estimation / Hypothesis Testing)

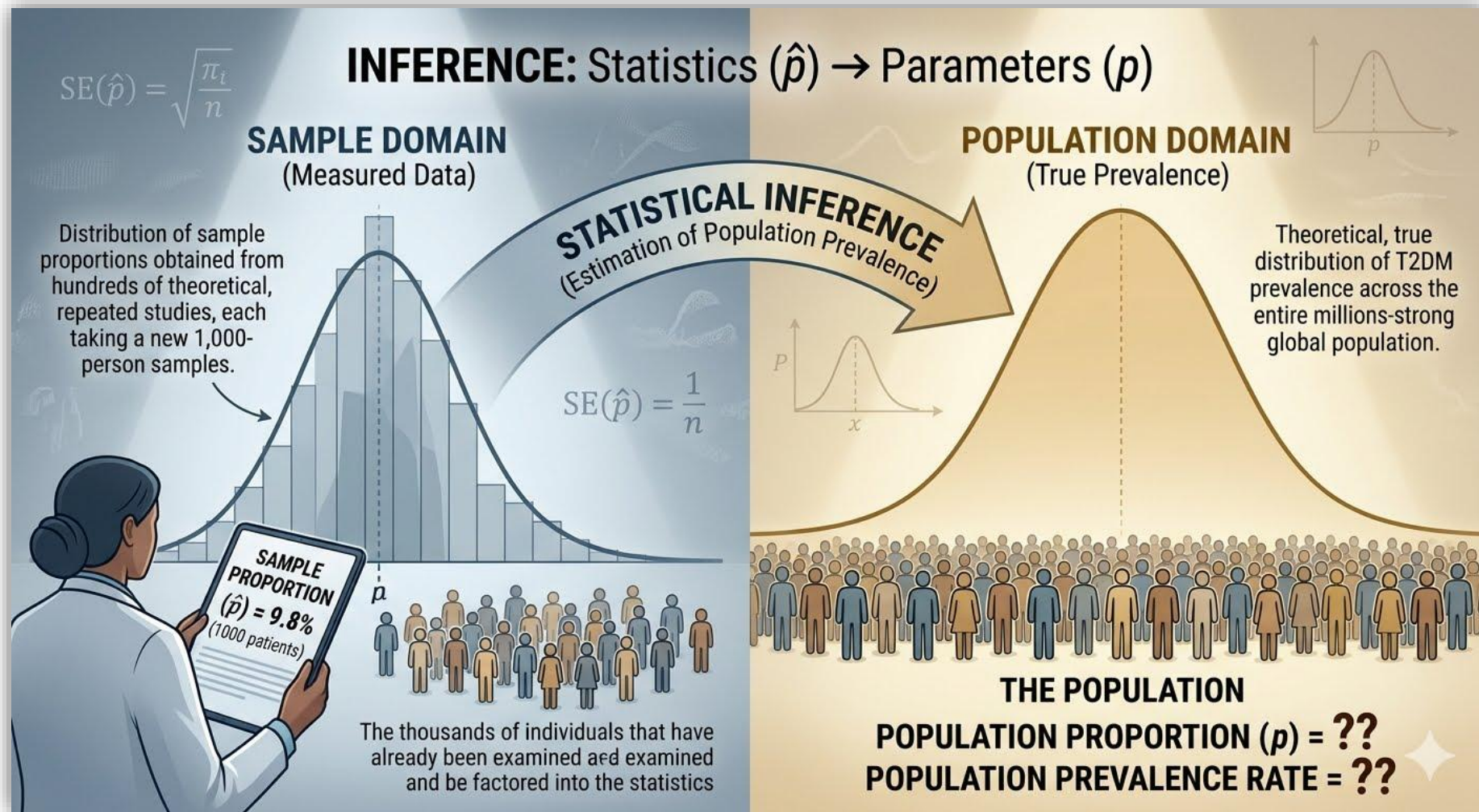
POPULATION DOMAIN

(N = millions)



TRUTH

POPULATION MEAN (μ) = ??
POPULATION STANDARD DEVIATION (σ) = ??



Standard Deviation (SD) vs Standard Error of Mean (SEM)



DOMAIN 1: INDIVIDUAL VARIABILITY (DATA DISTRIBUTION)

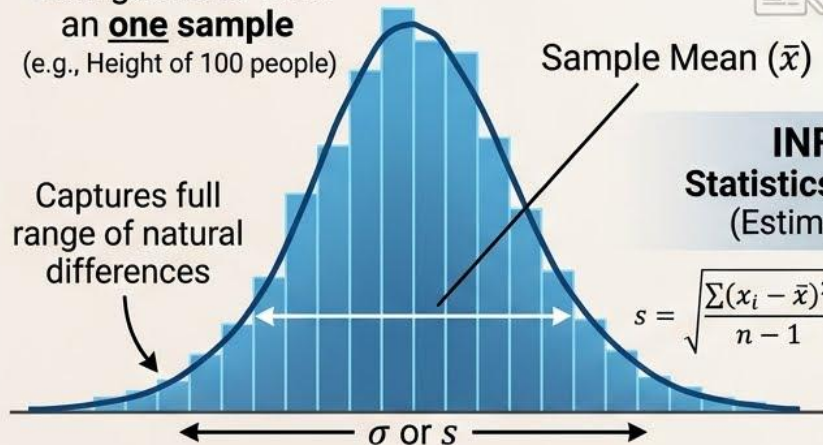


Data gathered from
an **one sample**
(e.g., Height of 100 people)

Captures full
range of natural
differences

Sample Mean ($\bar{x}$)

INFERENCE:
Statistics $\rightarrow$ Parameters
(Estimate Precision)



Standard Deviation (SD) measures how much
individual data points vary from their own sample
average.

The spread of individual numbers

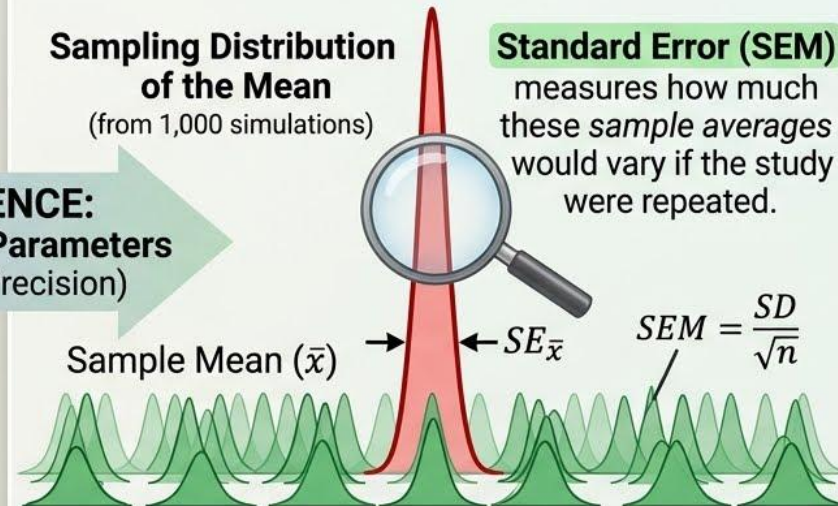


DOMAIN 2: SAMPLING VARIABILITY (DISTRIBUTION OF STATISTICS)



**Sampling Distribution
of the Mean**
(from 1,000 simulations)

Standard Error (SEM)
measures how much
these *sample averages*
would vary if the study
were repeated.



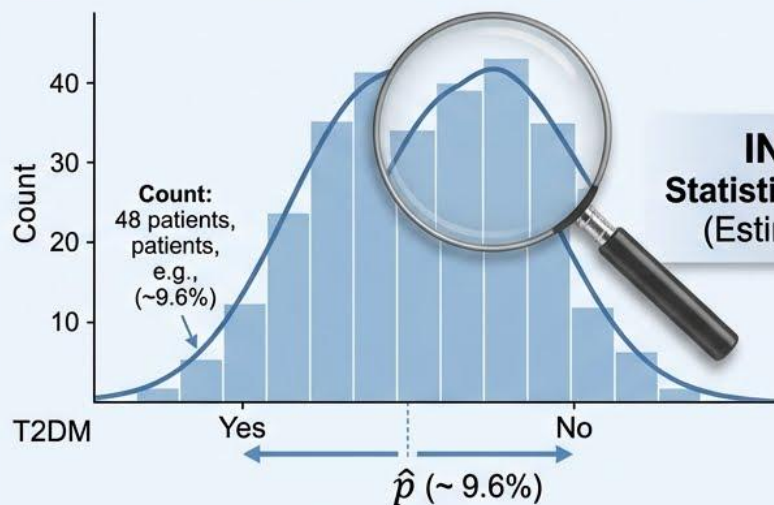
Standard Error (SEM) measures how much these
sample averages would vary if the study were repeated.

Precision of the estimated average

The Sample Variability vs. Sampling Variability (proportions)

DOMAIN 1: SAMPLE VARIABILITY (MEASURING INDIVIDUALS)

Distribution of individual survey responses (T2DM) gathered from one random sample of 500 people.

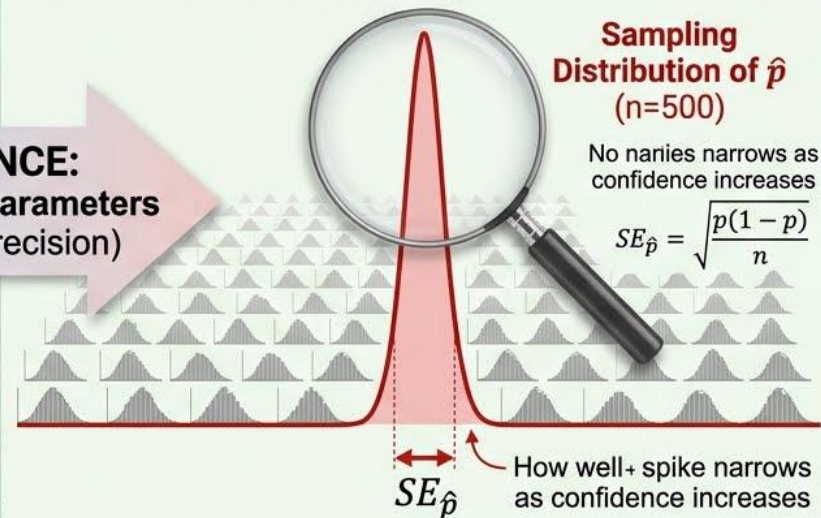


SAMPLE PROPORTION ($\hat{p}$) = measures the **natural difference** between individuals **within** this single dataset.

INFERENCE:
Statistics $\rightarrow$ Parameters
(Estimate Precision)

DOMAIN 2: SAMPLING VARIABILITY (PRECISION OF THE ESTIMATE)

Visual **distribution of statistics** (sample proportions) calculated if the study were repeated millions of times.

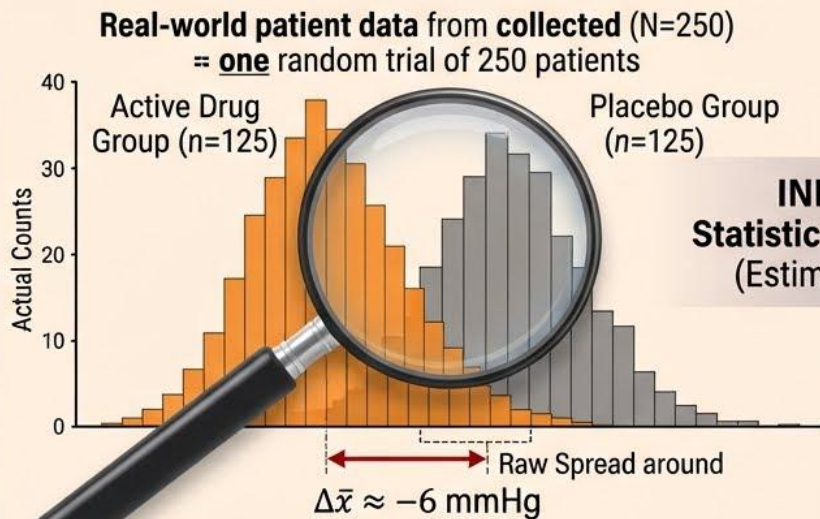


STANDARD ERROR (SE) = measures how much the **estimated proportion** would vary across many repeated studies

Sample Variability and. Sampling Variability

At the difference in means hypothetical clinical trial cohort, Active Drug vs. Placebo group
Measuring change in $\hat{\mu}$ vital continuous health outcome (e.g., Blood Pressure Reduction, -mmHg)

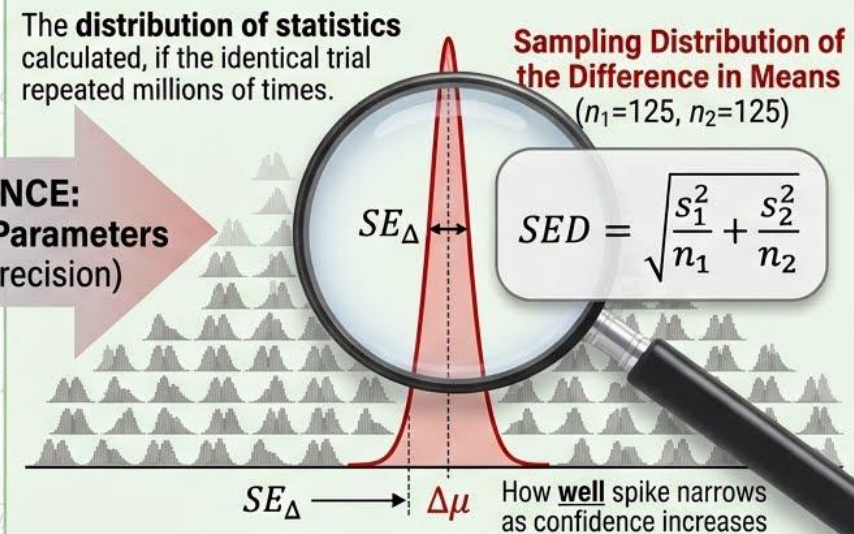
DOMAIN 1: SAMPLE VARIABILITY (MEASURING INDIVIDUALS)



SAMPLE DIFFERENCE ($\Delta \bar{x}$) =
measures the natural variation of individual
individual outcomes within this single trial.

INFERENCE:
Statistics $\rightarrow$ Parameters
(Estimate Precision)

DOMAIN 2: SAMPLING VARIABILITY (PRECISION OF THE ESTIMATE)



Standard Error of the Difference (SED)
measures how much the estimated clinical effect
would vary across many repeated trials.

One-Sample t-test

Purpose: Compare the mean of a single sample to a known population mean.

Hypothesis Example:

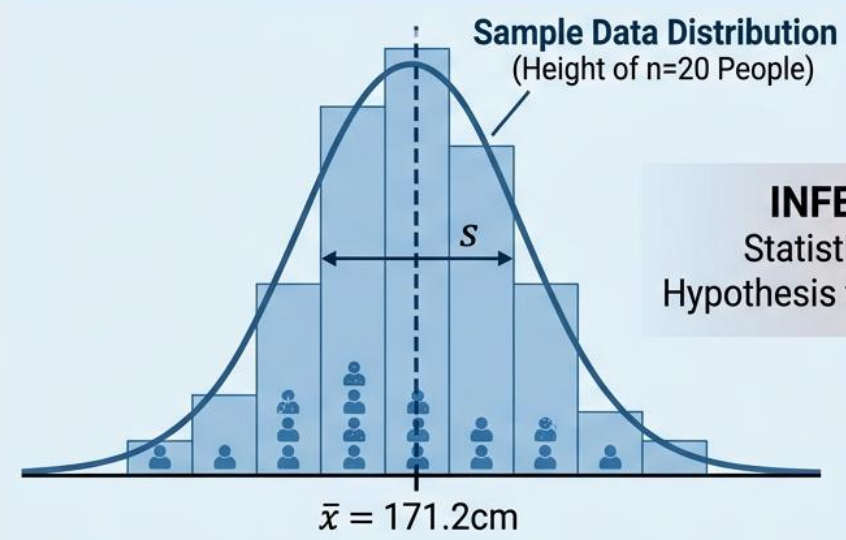
- H_0 : Sample mean = Population mean
- H_1 : Sample mean $\neq$ Population mean

When to Use:

- Continuous dependent variable (DV)
- Normally distributed data
- Single group compared to a known value

One-Sample T-test

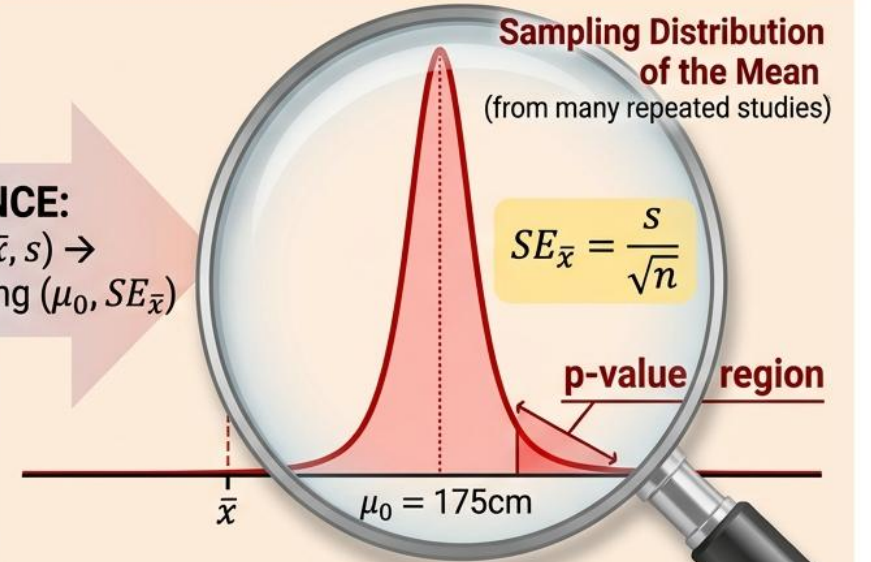
DOMAIN 1: INDIVIDUAL VARIABILITY (SAMPLE DATA)



SAMPLE VARIABILITY measures the natural differences between individ within this single sample.

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

DOMAIN 2: SAMPLING VARIABILITY (DISTRIBUTION OF STATISTICS)



STANDARD ERROR (SEM) measures how much the estimated average would fluctuate across millions of repeated studies.

INFERENCE:
Statistics $(\bar{x}, s) \rightarrow$
Hypothesis testing $(\mu_0, SE_{\bar{x}})$

+ ONE-SAMPLE T-TEST: STATISTICAL HYPOTHESIS & REJECTION REGION



Statistical Hypothesis (Two-Tailed): (One-Sample T)

Null hypothesis $H_0: \mu = \mu_0 = 100$ Alternative hypothesis $H_1: \mu \neq \mu_0$, where $\mu \neq 100$



Sample Data & Test Statistic:

$(x_i): \bullet, \bullet, \bullet, \dots, \bullet, \dots, \bullet, \dots$ $\bar{x} = 112$ $s = 20$ $n = 25$ $t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{112 - 100}{\frac{20}{\sqrt{25}}} = \frac{12}{4} = 3.0$

Bell Curve (T-Distribution)

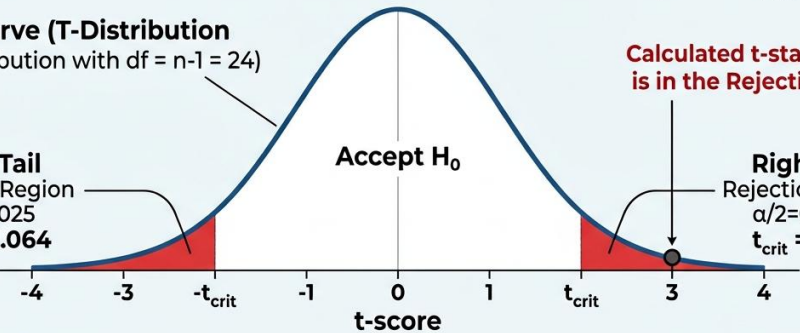
(T-Distribution with $df = n-1 = 24$)

Left Tail
Rejection Region
 $\alpha/2=0.025$
 $t_{crit} = -2.064$

Accept H_0

**Calculated t-statistic ($t = 3.0$)
is in the Rejection Region**

Right Tail
Rejection Region
 $\alpha/2=0.025$
 $t_{crit} = 2.064$



Decision Rule: Reject H_0 if $t < -2.064$ or $t > 2.064$.

+ ONE-SAMPLE T-TEST: ONE-TAILED STATISTICAL HYPOTHESIS & REJECTION REGION



Statistical Hypothesis (Right-Tailed Example)

Null hypothesis $H_0: \mu \leq \mu_0 = 100$ Alternative hypothesis $H_1: \mu > \mu_0$, where $\mu > 100$



Sample Data & Test Statistic:

$(x_i): \bullet, \bullet, \bullet, \dots, \bullet, \dots, \bullet, \dots$ $\bar{x} = 118$ $s = 22.5$ $n = 36$ $t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{118 - 100}{\frac{22.5}{\sqrt{36}}} = \frac{18}{3.75} = 4.8$

Bell Curve (T-Distribution)

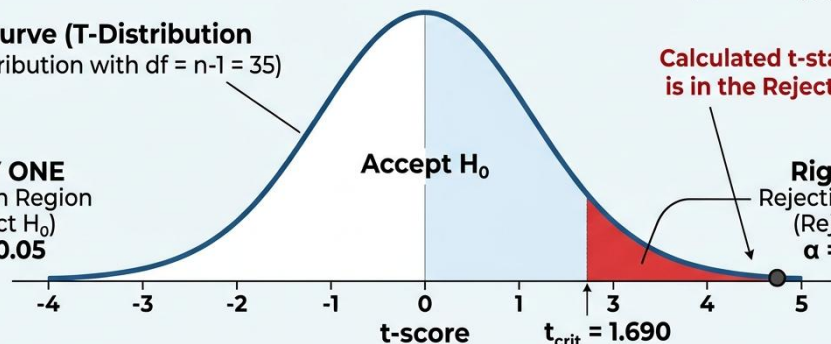
(T-Distribution with $df = n-1 = 35$)

ONLY ONE
Rejection Region
(Reject H_0)
 $\alpha = 0.05$

Accept H_0

**Calculated t-statistic ($t = 4.8$)
is in the Rejection Region**

Right Tail
Rejection Region
(Reject H_0)
 $\alpha = 0.05$



Decision Rule: Reject H_0 if $t > 1.690$. Fail to reject H_0 if $t \leq 1.690$.

Independent (Two-Sample) t-test

Purpose: Compare means between two independent groups.

Hypothesis Example:

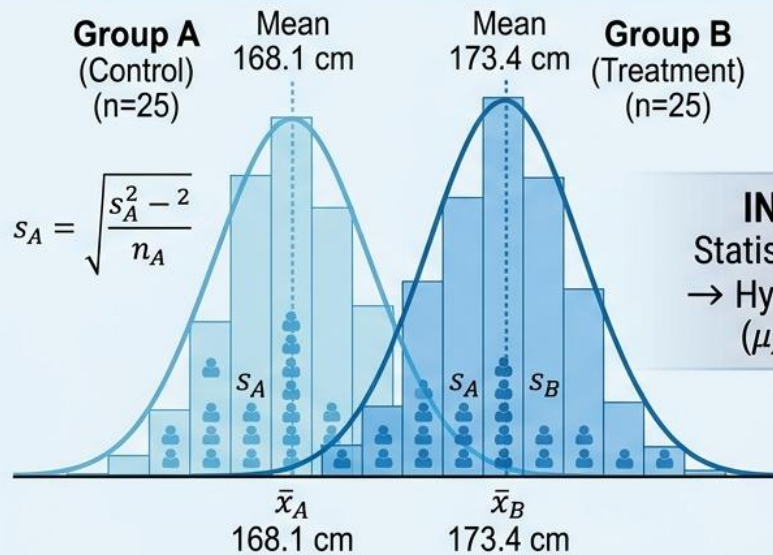
- H_0 : Group A mean = Group B mean
- H_1 : Group A mean $\neq$ Group B mean

When to Use:

- Continuous DV
- Two independent groups (e.g., control vs. treatment)
- Normally distributed data with equal variances

Independent T-test

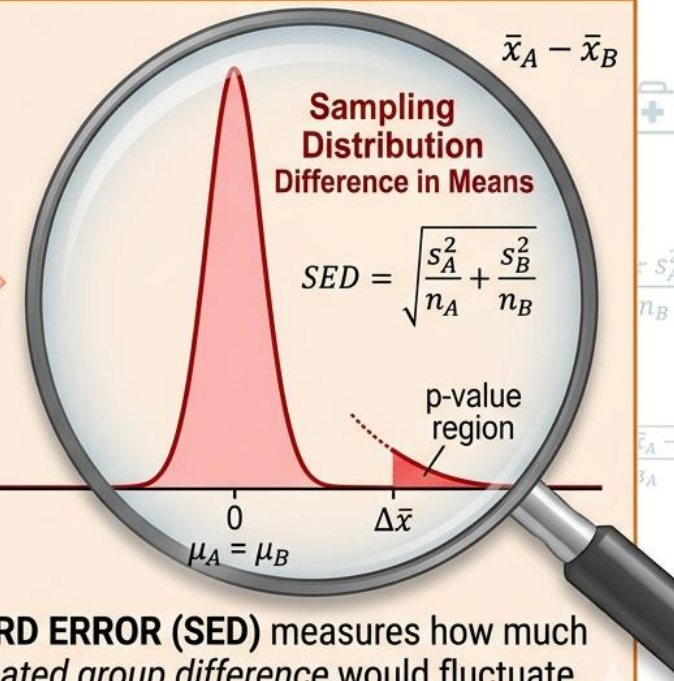
DOMAIN 1: INDIVIDUAL VARIABILITY (SAMPLE DATA)



INDIVIDUAL VARIABILITY measures how much single people vary within their own distinct treatment groups.

INFERENCE:
Statistics ($\bar{x}_A, S_A, S_B$)
→ Hypothesis testing
($\mu_A = \mu_B, SED$)

DOMAIN 2: SAMPLING VARIABILITY (DISTRIBUTION OF STATISTICS)



STANDARD ERROR (SED) measures how much the estimated group difference would fluctuate across millions of repeated identical trials.

Dependent (Paired-Sample) t-test

Purpose: Compare means from the same group at two different times (or matched pairs).

Hypothesis Example:

- H_0 : Mean difference (Time 1 – Time 2) = 0
- H_1 : Mean difference $\neq 0$

When to Use:

- Repeated measures (pre-test vs. post-test)
- Matched pairs (e.g., twins in different conditions)

Dependent (Paired-Samples) T-test

DOMAIN 1: INDIVIDUAL VARIABILITY

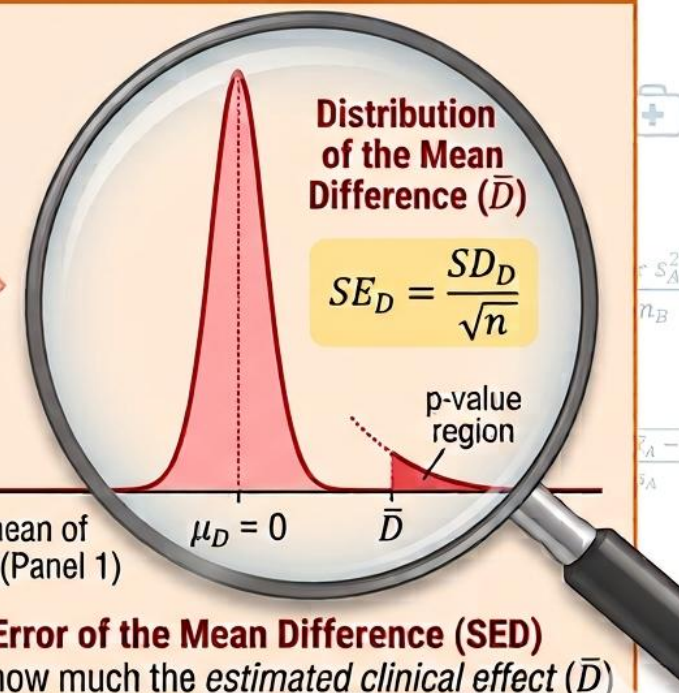
PAIRED TRIAL DATA: RAW INDIVIDUAL OUTCOMES (N=5 People)

Patient ID	Before Treatment (-mmHg)	After Treatment (-mmHg)	Difference (After-Before)
Patient 1	140	127	-6
Patient 2	108	123	-15
Patient 3	160	185	+25
Patient 4	136	175	+39
Patient 4	169	136	-33
Patient 5	159	122	-37

RAW INDIVIDUAL VARIABILITY is captured here; the t-test must manage these diverse changes to find a common signal.

INFERENCE:
Statistics ($\bar{x}_{\text{before}}, \bar{x}_{\text{after}}$)
→ Hypothesis testing
($\mu_D = 0, SED$)

DOMAIN 2: SAMPLING VARIABILITY (DISTRIBUTION OF STATISTICS)



Sample mean of differences (Panel 1)

Standard Error of the Mean Difference (SED) measures how much the estimated clinical effect ($\bar{D}$) would fluctuate across millions of repeated paired trials.

One-Way ANOVA (Analysis of Variance)

Purpose: Compare means among three or more independent groups.

Hypothesis Example:

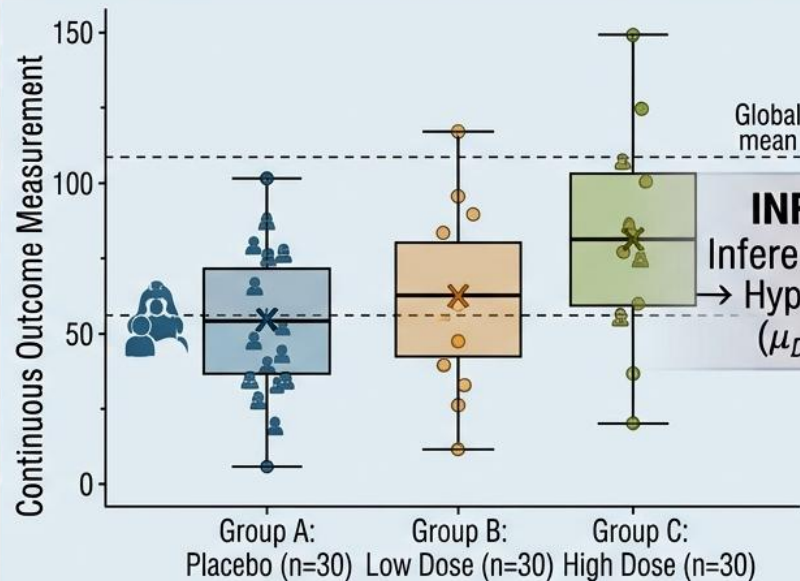
- H_0 : All group means are equal
- H_1 : At least one group mean differs

When to Use:

- One categorical IV (independent variable) with ≥ 3 levels
- Continuous DV
- Normally distributed data with homogeneity of variance

Understanding One-Way ANOVA: Testing Group Means

DOMAIN 1: VISUALIZING RAW INDIVIDUAL DATA

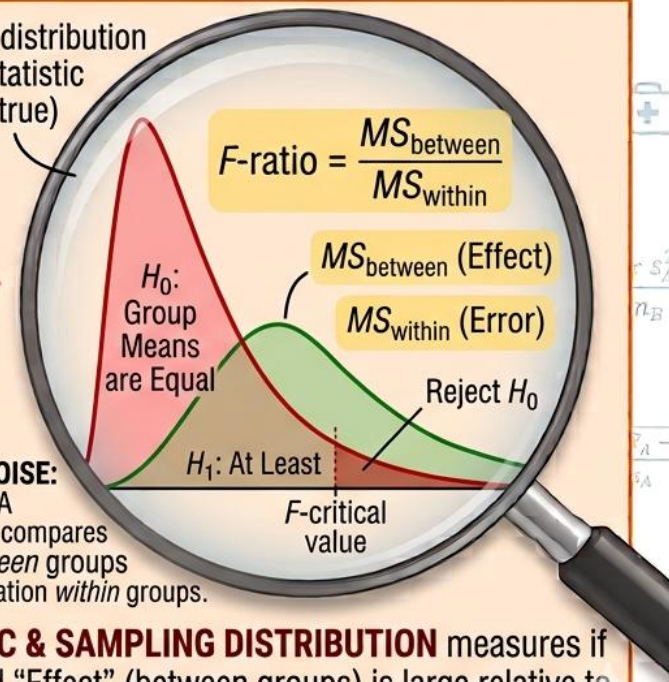


INDIVIDUAL VARIABILITY: Does medicine type significantly change average results *above and beyond* individual noise?

INFERENCE:
Inferential statistics
→ Hypothesis testing
($\mu_D = 0, SED$)

DOMAIN 2: MEASURING SAMPLING VARIABILITY & PRECISION

The sampling distribution of the F-statistic (if H_0 is true)



SIGNAL-TO-NOISE:
One-way ANOVA mathematically compares variation *between* groups against the variation *within* groups.

F-STATISTIC & SAMPLING DISTRIBUTION measures if the observed "Effect" (between groups) is large relative to random "Error" (within groups) across repeated studies.

Two-Way ANOVA

Purpose: Examine the effect of two independent categorical variables on a continuous DV, including interaction effects.

Hypothesis Example:

- H_{01} : No effect of Factor A
- H_{02} : No effect of Factor B
- H_{03} : No interaction between Factor A and B

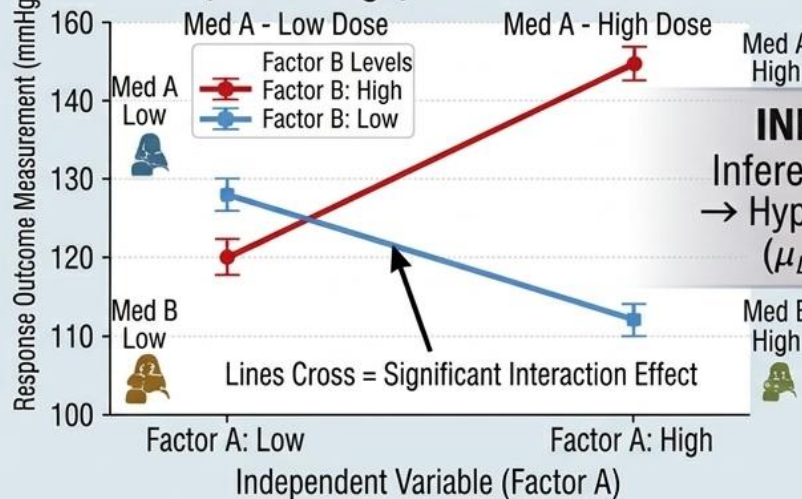
When to Use:

- Two categorical IVs
- Tests main effects and interaction effects

Understanding 2-Way ANOVA: Testing Factorial Effects & Interaction

DOMAIN 1: VISUALIZING RAW INDIVIDUAL DATA - 2x2 FACTORIAL DESIGN

Effect of Medicine Type (A vs B) & Dose (Low vs High) on Blood Pressure

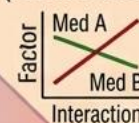


INDIVIDUAL VARIABILITY: Does medicine effect depend on dosage *above and beyond* individual noise? We analyze variances *between* the four combinations.

INFERENCE:
Inferential statistics
→ Hypothesis testing
($\mu_D = 0$, SED)

DOMAIN 2: MEASURING SAMPLING VARIABILITY & PRECISION - INTERACTION ANALYSIS

Null Sampling Distribution (No Interaction)



$F_{\text{interaction_distribution (If Null)}}$

$$F\text{-ratio} = \frac{MS_{\text{interaction}}}{MS_{\text{within}}}$$

$F_{\text{interaction_distribution (Observed)}}$

MS_{between} (Main Factors & Interaction)

MS_{within} (Error)

Observed Sampling Distribution (With Significant Interaction)

Reject H_0 (Reject No Interaction)

Reject H_0 Interaction)

F-critical value

FACTORIAL SIGNAL-TO-NOISE:

We test if the "Interaction" (Factor A * Factor B) explains significant variance over the "Main Effects" and standard individual "Error".

F-STATISTIC & SAMPLING DISTRIBUTION: We analyze multiple F-ratios (Main A, Main B, Interaction) against random "Error". If the observed "Interaction" is large relative to random across repeated that, we conclude an interaction effect.

Repeated Measures ANOVA

Purpose: Compare means of the same group across multiple time points (or conditions).

Hypothesis Example:

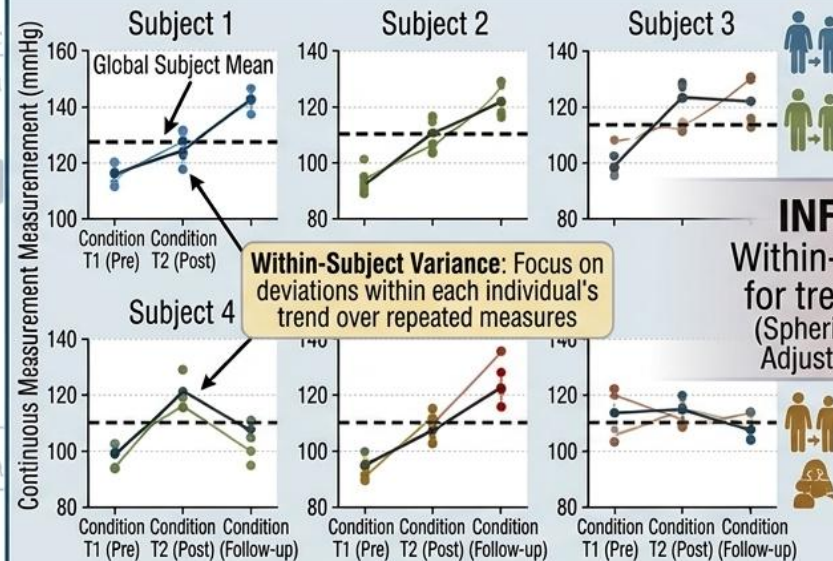
- H_0 : No difference across time points
- H_1 : At least one time point differs

When to Use:

- Within-subjects design (same participants measured repeatedly)
- More than two time points (otherwise, use paired t-test)

Understanding Repeated Measures ANOVA: Testing Within-Subject Trends & Sphericity

DOMAIN 1: VISUALIZING WITHIN-SUBJECT CHANGE – Longitudinal Profile Design



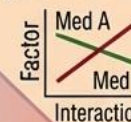
Within-Subject Variance: Focus on deviations within each individual's trend over repeated measures

WITHIN-SUBJECT VARIABILITY: Does measurement significantly change across repeated conditions for the same individual, accounting for their unique starting point? We analyze variances *across* the condition means for each subject.

INFERENCE:
Within-Subject F-test for treatment effect (Sphericity Assumed or Adjusted, e.g., GG/HF)

DOMAIN 2: MEASURING SYSTEMATIC WITHIN-SUBJECT VARIANCE & PRECISION - WITHIN-GROUP COMPARISONS

Null Sampling Distribution (No Interaction)



$F_{ws-ratio_distribution} \text{ (If Null)}$

$$F_{ws-ratio} = \frac{MS_{\text{between_Repeated_Conditions}}}{MS_{\text{Error_Within_Subject}}}$$

$F_{ws-ratio_distribution} \text{ (Observed)}$

Signal (Systematic Treatment Effect) - to - Noise (Random Subject Residual Deviation)

$F_{ws-ratio} \text{ (Factor Repeated)}$

SPHERICITY SIGNAL-TO-NOISE:

We analyze variances *between* repeated conditions for the *same individuals*, which often yields greater statistical power.

F-STATISTIC & WITHIN-SUBJECT DISTRIBUTIONS: We analyze multiple F-ratios (Repeated Condition Main Effect, Subject variability, etc.) against specific within-subject error terms. Sphericity testing (e.g., Mauchly's). If the within-subject signal is large relative to specific residual variability, we conclude a time/condition effect.

Summary of Statistical Test for Means

Test	Purpose	IV & DV	Groups/Time Points
One-Sample t-test	Compare sample mean to population mean	1 continuous DV	1 group vs. known value
Independent t-test	Compare two independent groups	1 categorical IV (2 levels) + 1 continuous DV	2 groups
Dependent t-test	Compare two related measurements (paired/matched)	1 continuous DV	2 time points/matched pairs
One-Way ANOVA	Compare ≥ 3 independent groups	1 categorical IV (≥ 3 levels) + 1 continuous DV	≥ 3 groups
Two-Way ANOVA	Test two IVs + interaction effects	2 categorical IVs + 1 continuous DV	Multiple groups
Repeated Measures ANOVA	Compare same group across ≥ 3 time points	1 within-subjects IV (≥ 3 levels) + 1 continuous DV	≥ 3 repeated measures

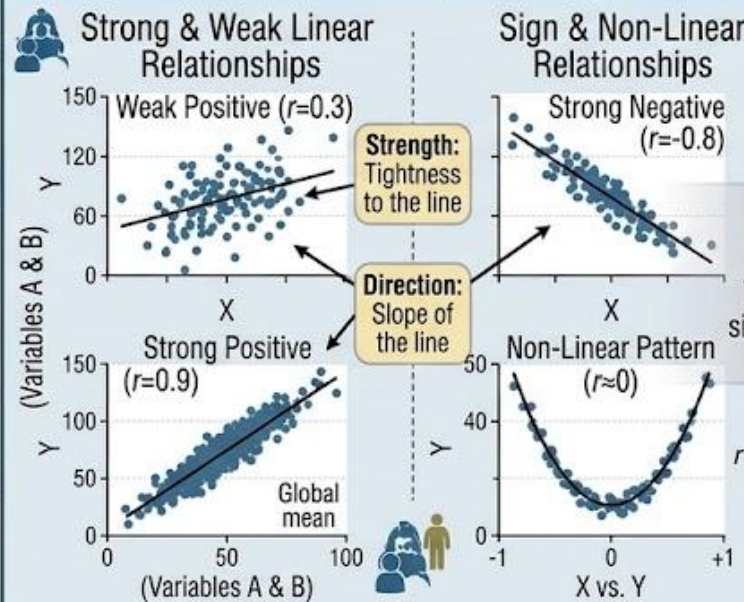
Correlation analysis

Purpose: Measures the strength and direction of association between two variables.

Correlation Test	Variable Types	Hypothesis	When to Use
Pearson's r	Two continuous variables	$H_0: \rho = 0$ (no correlation) $H_1: \rho \neq 0$ (correlation exists)	Linear relationship, normally distributed data
Spearman's ρ (rho)	Two ordinal/ranked or non-normal continuous variables	H_0 : No monotonic relationship H_1 : Monotonic relationship exists	Non-linear but monotonic trends
Kendall's τ (tau)	Two ordinal/ranked variables (small sample sizes)	H_0 : No association H_1 : Association exists	Better than Spearman for tied ranks
Point-Biserial	One continuous + one binary variable	H_0 : No correlation H_1 : Correlation exists	E.g., Exam score (continuous) vs. Pass/Fail (binary)
Phi Coefficient	Two binary variables	H_0 : No association H_1 : Association exists	2x2 contingency tables
Cramer's V	Two categorical variables	H_0 : No association H_1 : Association exists	Larger contingency tables

Understanding Correlation Analysis: Testing Strength & Direction of Linear Relationships

DOMAIN 1: VISUALIZING BIVARIATE DATA – Scatterplot Designs

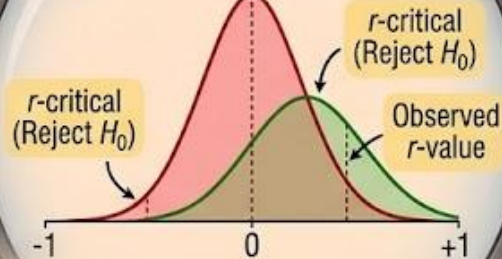


BIVARIATE VARIABILITY: How much of the variance in one variable is shared with the other? Visual inspection is essential before calculation.

DOMAIN 2: MEASURING CORRELATION COEFFICIENTS & PRECISION – r -value distributions

Null Sampling Distribution ($r=0$)

Sampling Distribution of r (If H_0 is true)



INFERENCE:

Statistical Inference: Testing if Pearson's r is significantly different from zero ($H_0: \rho=0, H_1: \rho \neq 0$)

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

r-STATISTIC & NULL DISTRIBUTIONS: We analyze if our observed correlation (r) is statistically probable if there were zero correlation in the population. We test r against its standard error ($SE_r = \sqrt{\frac{1-r^2}{n-2}}$) with $n-2$ degrees of freedom.

PRECISION & CONFIDENCE INTERVALS: We can construct a 95% CI for ρ , showing the range of plausible true correlation values, influenced by sample size and r -value magnitude.

Simple Linear Regression

Purpose: Predicts a continuous dependent variable (DV) from one continuous independent variable (IV).

Model:

- $Y = \beta_0 + \beta_1 X_1 + \epsilon$

Hypothesis Testing:

- Overall Model (F-test):
 - H_0 : The model does not fit the data better than the mean
 - H_1 : The model fits the data better
- Coefficient (t-test):
 - H_0 : $\beta_1 = 0$ (no effect of X on Y)
 - H_1 : $\beta_1 \neq 0$ (X affects Y)

Assumptions:

- Linearity
- Independence of residuals
- Homoscedasticity (constant variance)
- Normality of residuals

Multiple Linear Regression

Purpose: Predicts a continuous DV from two or more IVs (continuous/categorical).

Model:

- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$

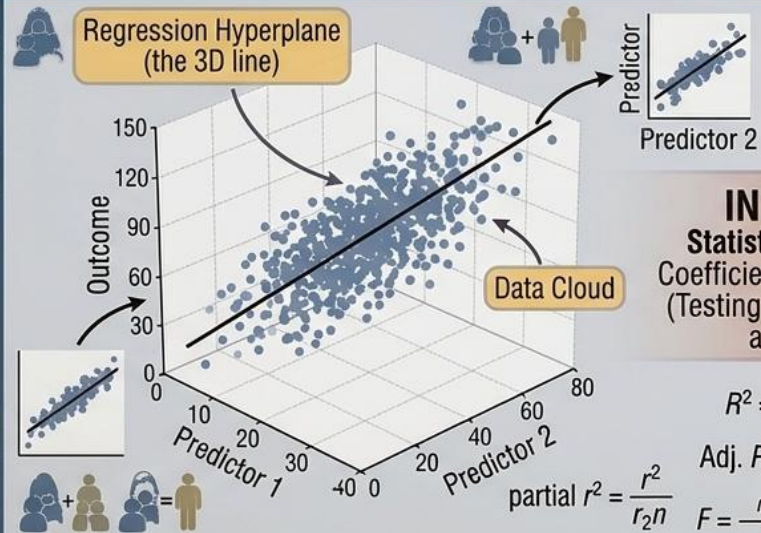
Hypothesis Testing:

- Overall Model (F-test):
 - H_0 : All slopes ($\beta_1, \beta_2, \dots, \beta_k$) = 0
 - H_1 : At least one slope $\neq 0$
- Individual Coefficients (t-tests):
 - H_0 : $\beta_i = 0$ (no effect of X_i on Y)
 - H_1 : $\beta_i \neq 0$ (X_i affects Y)

Assumptions: Same as simple regression + no multicollinearity (check with VIF).

Understanding Multiple Regression Analysis: Predicting Outcomes with Multiple Independent Variables

DOMAIN 1: VISUALIZING MULTI-VARIABLE DATA – Hyperplane & Predictor Interaction



MULTIVARIATE VARIABILITY: We explore how changes in multiple predictors jointly relate to an outcome. Standard (β) are used to compare the relative strength of influence. (e.g., $\beta_1=0.7$, $\beta_2=-0.4$)

LEGEND: Variable X_k - Predictor k ; Variable Y - Outcome; β - Standardized Coefficient; b - Unstandardized Coefficient; R^2 - Coefficient of Determination; Sampling Distribution; H_0 - Null Hypothesis; Reject H_0

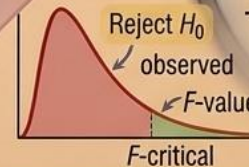
DOMAIN 2: MEASURING COEFFICIENTS & MODEL FIT – R^2 & ANOVA distributions

Null Sampling Distributions for b -coefficients

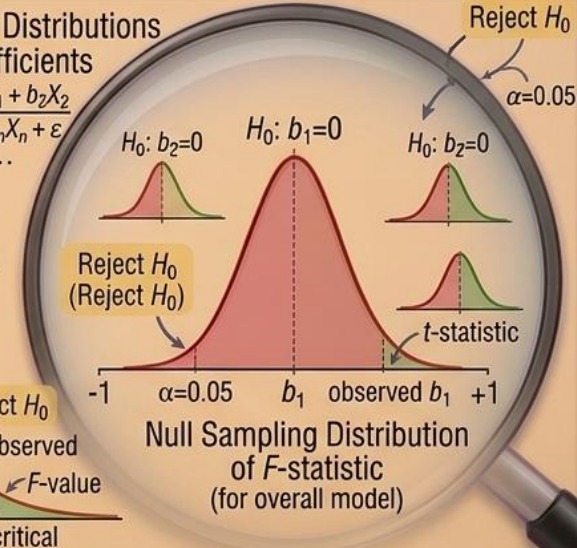
$$Y = \frac{b_0 + b_1X_1 + b_2X_2}{+ \dots + b_nX_n + \varepsilon}$$

$$SE_{b_k} = \dots$$

INFERENCE:
Statistical Inference of Coefficients & Overall Model
(Testing $H_0: b_1=b_2=\dots=b_n=0$, and $H_0: b_k=0$)



F-TEST & MODEL FIT: We test if the entire set of predictors explains a significant portion of variance in Y compared to a null model (Y -intercept only). $F = MS_{\text{regression}}/MS_{\text{error}}$



Logistic Regression

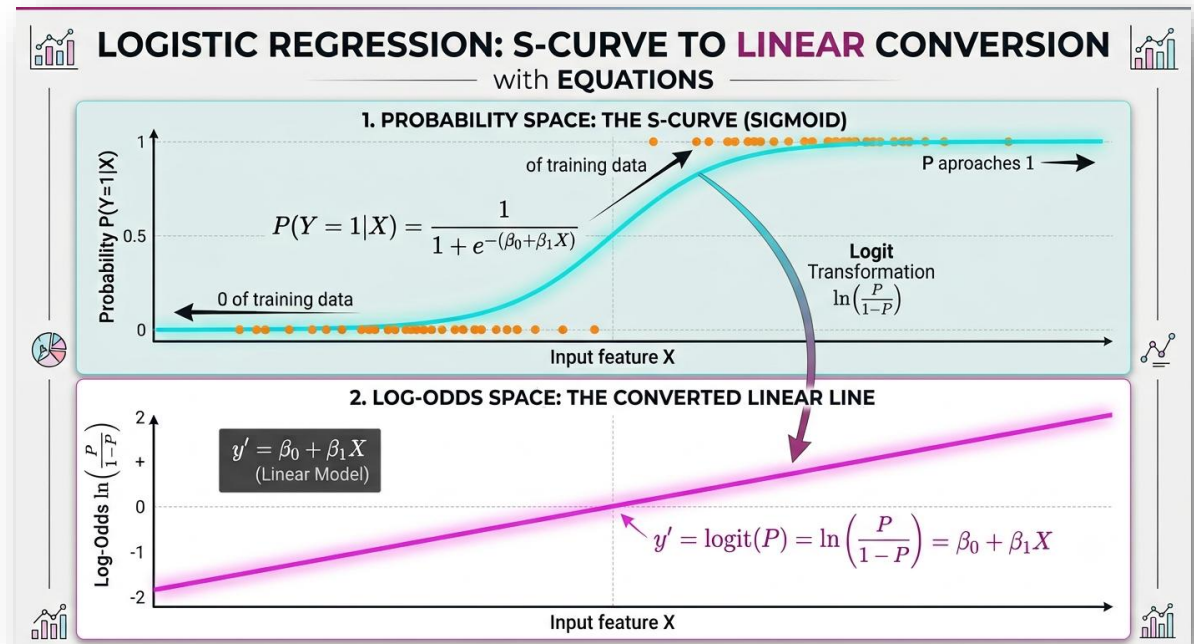
Purpose: Predicts a binary/categorical DV (e.g., Yes/No, Success/Failure) from one or more IVs.

Types:

- Binary Logistic Regression (DV has 2 categories)
- Multinomial Logistic Regression (DV has >2 unordered categories)
- Ordinal Logistic Regression (DV has ordered categories, e.g., Low/Medium/High)

Model (Binary):

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$



Hypothesis Testing:

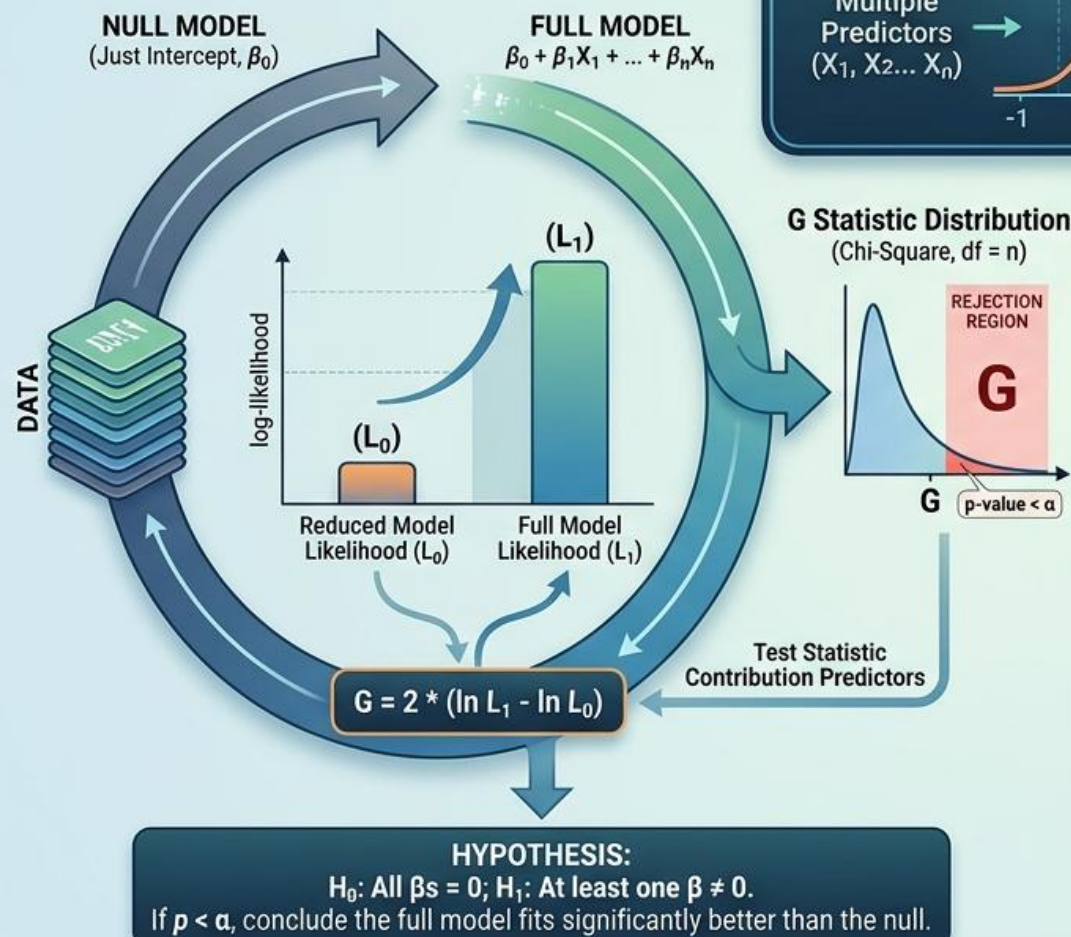
- Overall Model (Likelihood Ratio Test):
 - H_0 : All $\beta_i=0$ (model with no predictors fits as well)
 - H_1 : At least one $\beta_i \neq 0$
- Wald Test (Individual Coefficients):
 - H_0 : $\beta_i=0$ (no effect)
 - H_1 : $\beta_i \neq 0$ (significant effect)

Assumptions:

- ✓ No multicollinearity
- ✓ Linearity between log-odds & IVs
- ✓ Large sample size (~10 events per predictor)

PART 1: OVERALL MODEL SIGNIFICANCE TEST

A: Likelihood Ratio Test (Overall)



STATISTICAL TESTS FOR LOGISTIC REGRESSION: MODEL & INDIVIDUAL PREDICTORS

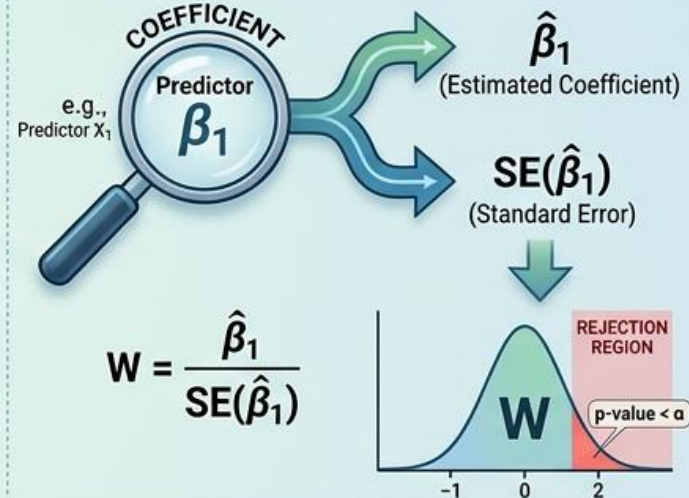
Multiple Predictors ($X_1, X_2, \dots, X_n$)

Categorical Outcome (Y)



PART 2: SIGNIFICANCE TEST FOR EACH PREDICTOR

B: Wald Test (Individual Predictors)



INDIVIDUAL PREDICTOR:
 $H_0: \beta = 0$; $H_1: \beta \neq 0$. If $p < \alpha$, conclude the predictor has a significant unique effect on outcome odds.

C: INTEGRATED SUMMARY TABLE (Example Output)

Test	Hypothesis (H_0)	Statistic	Distribution	p-value	Decision	Visualization
Model Significance (LRT)	H_0 : All $\beta_s = 0$;	G		$p < 0.001$	Conclude	
Predictor X1 (Wald)	H_0 : $\beta = 0$;	$SE(\hat{\beta}_1)$		0.028	Conclude	
Predictor X2 (Wald)	H_1 : $\beta \neq 0$;	$SE(\hat{\beta}_1)$		0.031	Conclude	

α (Significance Level)
 df (Degrees of Freedom)
Unique Effect

Summary for Models

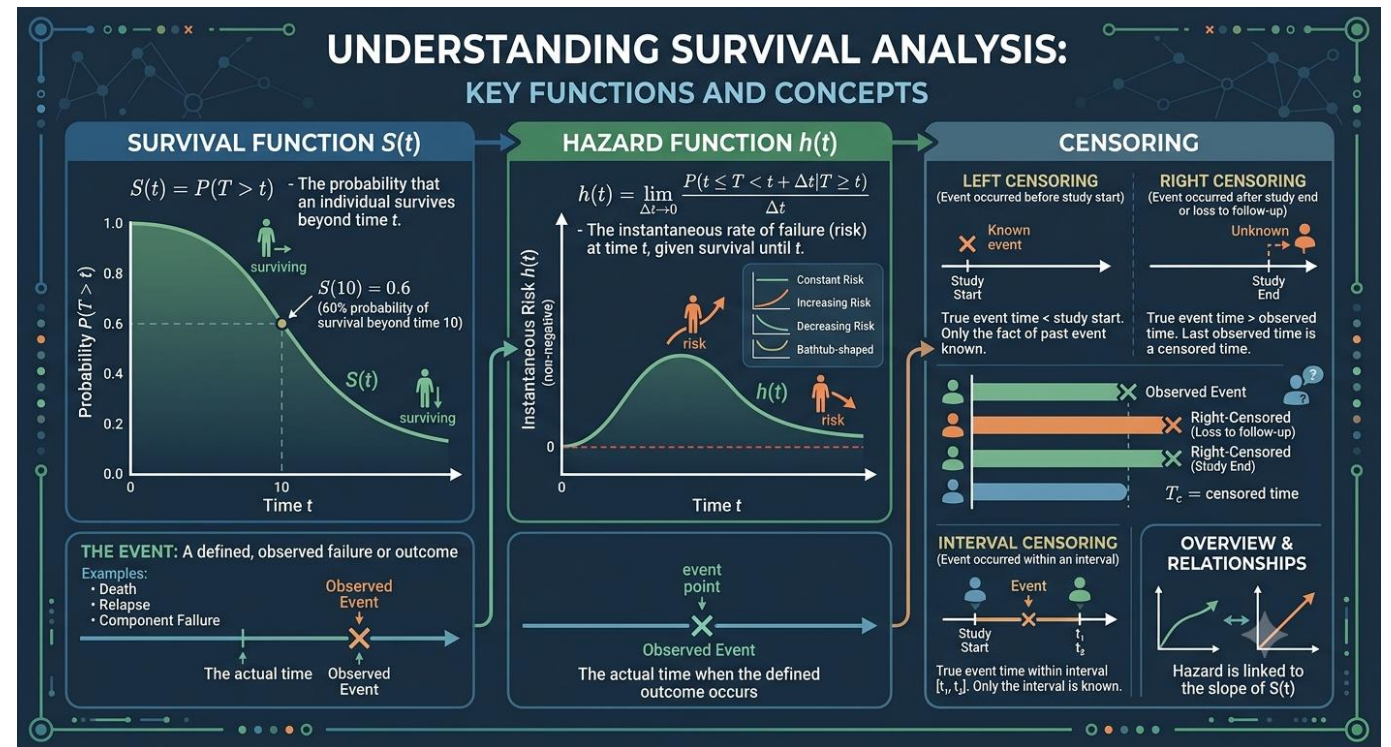
Model	DV Type	IV Type	Key Hypothesis Tests
Simple Regression	Continuous	1 Continuous	F-test (model), t-test (slope)
Multiple Regression	Continuous	≥ 2 Continuous/Categorical	F-test (model), t-tests (slopes)
Binary Logistic	Binary (0/1)	≥ 1 Continuous/Categorical	Likelihood Ratio, Wald Test
Multinomial Logistic	Unordered (>2 categories)	≥ 1 Continuous/Categorical	Likelihood Ratio, Wald Test
Ordinal Logistic	Ordered Categories	≥ 1 Continuous/Categorical	Likelihood Ratio, Wald Test

Survival Analysis

Purpose: Analyze **time-to-event** data (e.g., death, failure, recovery).

Key Concepts:

- Event: The outcome of interest (e.g., patient relapse).
- Censoring: Incomplete observation (e.g., study ends before event occurs).
- Survival Function ($S(t)$): Probability of surviving past time t .
- Hazard Function ($\lambda(t)$): Instantaneous risk of event at time t .



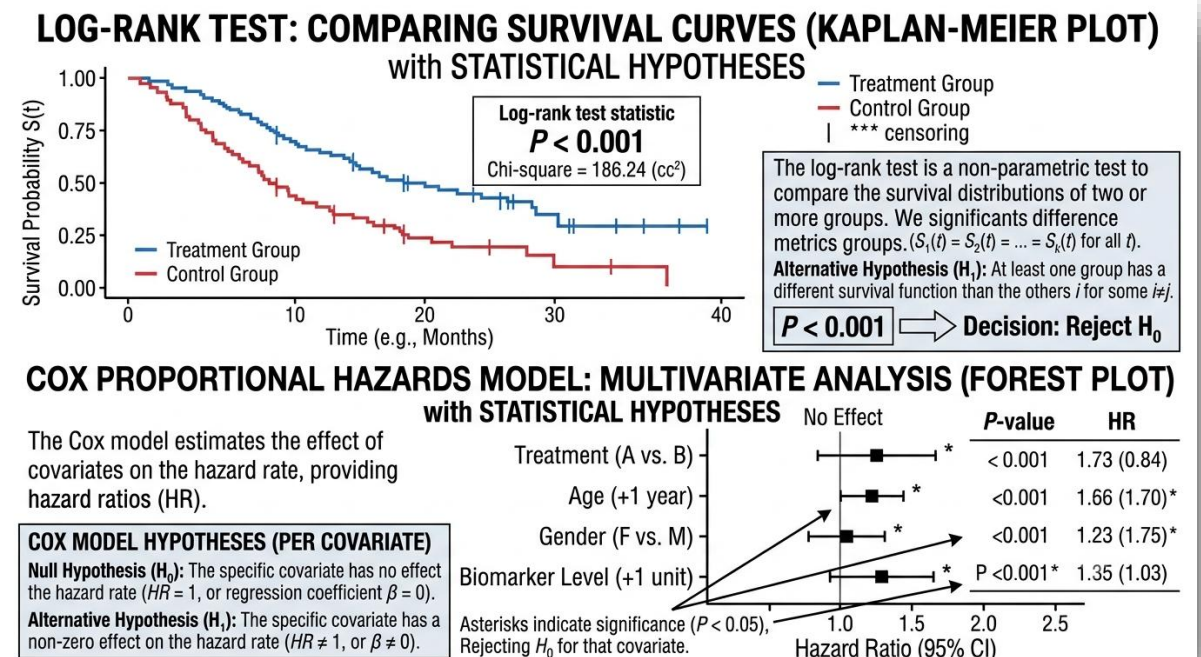
Hypothesis Testing in Survival Analysis

Examples:

- Log-Rank Test:
 - H_0 : Survival curves are identical across groups.
 - H_1 : Survival differs between groups.
 - Test Statistic: Chi-square.
- Cox Model (Wald Test):
 - H_0 : Hazard ratio (HR) = 1 (no effect).
 - H_1 : $HR \neq 1$ (predictor affects survival).

Assumptions:

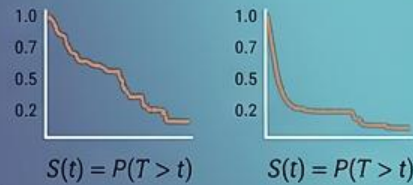
- Proportional hazards (Cox model).
- Independent censoring.



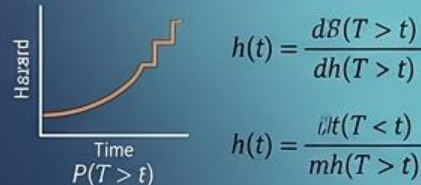
SURVIVAL ANALYSIS: A COMPLETE OVERVIEW

KEY CONCEPTS

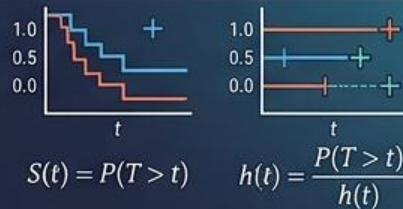
SURVIVAL FUNCTION $S(t)$



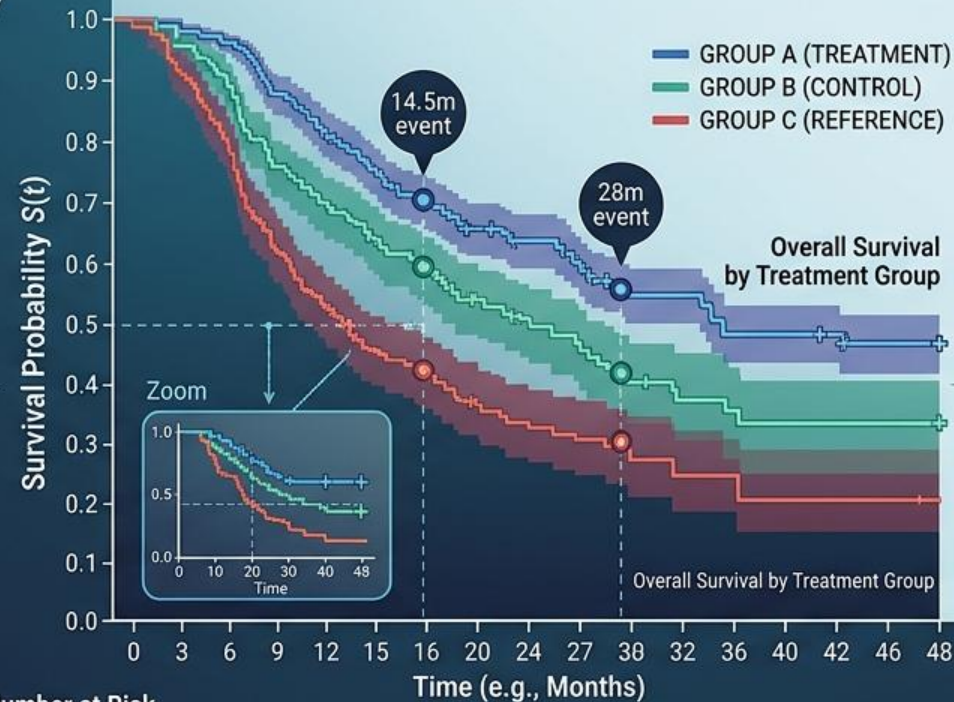
HAZARD FUNCTION $h(t)$



CENSORING (Left, Right, Interval)



PROBABILITY OF SURVIVAL (Kaplan-Meier)

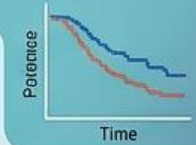


Number at Risk

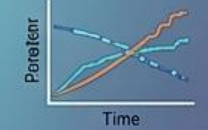
	0	3	6	9	12	15	16	20	24	27	30	32	36	40	42	46	48
GROUP A (TREATMENT)	100	114	115	130	68	32	28	15	0								
GROUP B (CONTROL)	102	80	86	74	28	24	13	2	0								
GROUP C (REFERENCE)	80	93	68	36	28	24	28	0	0								

MODELS & ANALYSIS

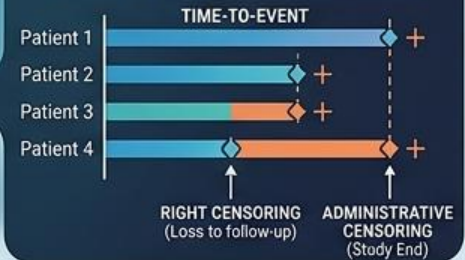
KAPLAN-MEIER (Non-Parametric)



COX PROPORTIONAL HAZARDS (Semi-Parametric)



INDIVIDUAL TRAJECTORIES



APPLICATIONS



Type	Method	Assumptions	Hypothesis Tests	Strengths
Non-Parametric	Kaplan-Meier	None	Log-Rank Test	Handles censoring, no distribution
Semiparametric	Cox Model	Proportional hazards	Wald Test, PH tests	Covariate adjustment
Parametric	Weibull	Specified distribution	Likelihood Ratio, AIC/BIC	Predicts beyond observed data
Descriptive	Life Table	Fixed intervals	N/A	Simplifies large datasets

Method	Key Feature	Hypothesis Testing	When to Use
Non-parametric			
Kaplan-Meier	Stepwise survival curve with censoring	Log-Rank Test (compare groups)	Small samples, exploratory analysis
Nelson-Aalen	Estimates cumulative hazard $H(t)$	N/A (descriptive)	Rare events, small datasets
Semiparametric			
Cox PH Model	Models covariates without baseline hazard	Wald Test ($H_0: \beta=0$)	Multivariable analysis, PH assumption
Descriptive			
Life Table	Grouped survival probabilities	N/A (descriptive)	Actuarial science, large cohorts
Hazard Plot	Visualizes instantaneous risk	N/A (exploratory)	Identify risk patterns over time

Distribution	Hazard Shape	Hypothesis Testing	When to Use
Parametric			
Exponential	Constant hazard	Likelihood Ratio Test	Electronics failure, stable risk
Weibull	Flexible (inc/dec)	AIC/BIC for model selection	Clinical trials, reliability testing
Log-Normal	Peaked hazard	Compare nested models	Disease progression studies

```
. ttest age == 50
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
age	10,839	54.60107	.1106773	11.52267	54.38412	54.81802

mean = mean(age) t = 41.5720
H0: mean = 50 Degrees of freedom = 10838

Ha: mean < 50
Pr(T < t) = 1.0000

Ha: mean != 50
Pr(|T| > |t|) = 0.0000

Ha: mean > 50
Pr(T > t) = 0.0000

```
. ttest age == 54
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
age	10,839	54.60107	.1106773	11.52267	54.38412	54.81802

mean = mean(age) t = 5.4308
H0: mean = 54 Degrees of freedom = 10838

Ha: mean < 54
Pr(T < t) = 1.0000

Ha: mean != 54
Pr(|T| > |t|) = 0.0000

Ha: mean > 54
Pr(T > t) = 0.0000

```

> print(tidy_result1)
  estimate statistic p.value parameter conf.low conf.high      method alternative
1  56.3885  306.1548      0      75413 56.28358  56.49342 One Sample t-test  two.sided
> print(tidy_result3)
  estimate statistic    p.value parameter conf.low conf.high      method alternative
1  56.3885  1.653226 0.09828897      75413 56.28358  56.49342 One Sample t-test  two.sided
> View(tidy_result1)

```

```
. anova fbsnew agegroup
```

```

      Number of obs =    10,471    R-squared      =    0.0142
      Root MSE      =    62.0969    Adj R-squared =    0.0140

```

Source	Partial SS	df	MS	F	Prob>F
Model	579510.5	2	289755.25	75.14	0.0000
agegroup	579510.5	2	289755.25	75.14	0.0000
Residual	40364839	10,468	3856.0221		
Total	40944350	10,470	3910.6351		

```

> print(anovareult2)
Call:
  aov(formula = sbp2 ~ age_group)

Terms:
              age_group Residuals
Sum of Squares    26443.9 1733479.2
Deg. of Freedom         7      3622

Residual standard error: 21.87687
Estimated effects may be unbalanced

```

```
. ttest age == 60
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
age	10,839	54.60107	.1106773	11.52267	54.38412	54.81802

mean = mean(age)

t = -48.7808

H0: mean = 60

Degrees of freedom = 10838

Ha: mean < 60

Pr(T < t) = 0.0000

Ha: mean != 60

Pr(|T| > |t|) = 0.0000

Ha: mean > 60

Pr(T > t) = 1.0000

. regress fbsnew age

Source	SS	df	MS	Number of obs	=	10,471
Model	634582.074	1	634582.074	F(1, 10469)	=	164.81
Residual	40309767.8	10,469	3850.39334	Prob > F	=	0.0000
				R-squared	=	0.0155
				Adj R-squared	=	0.0154
Total	40944349.9	10,470	3910.63514	Root MSE	=	62.052

fbsnew	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	-.6784317	.0528463	-12.84	0.000	-.7820206	-.5748429
_cons	195.0764	2.943397	66.28	0.000	189.3068	200.846

```
. regress fbsnew age sex
```

Source	SS	df	MS	Number of obs	=	10,471
Model	704249.259	2	352124.63	F(2, 10468)	=	91.60
Residual	40240100.7	10,468	3844.10591	Prob > F	=	0.0000
				R-squared	=	0.0172
				Adj R-squared	=	0.0170
Total	40944349.9	10,470	3910.63514	Root MSE	=	62.001

fbsnew	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	-.7024947	.0531048	-13.23	0.000	-.8065903	-.5983991
sex	5.375164	1.262628	4.26	0.000	2.900173	7.850155
_cons	194.4028	2.945246	66.01	0.000	188.6295	200.176

Statistic for Grad.R* x linear_model x NOD_Homework x					
Filter Search Sum					
	term	estimate	std.error	statistic	p.value
	All	All	All	All	All
1	(Intercept)	148.3620047	3.14351283	47.196246	0.000000000000
2	ageht	0.1404324	0.03085919	4.550747	0.000005532889
3	bw2	0.0471433	0.03258575	1.446746	0.148060447755

```
. logistic low age lwt i.race smoke ptl ht ui
```

Logistic regression

Number of obs = 189

LR chi2(8) = 33.22

Prob > chi2 = 0.0001

Pseudo R2 = 0.1416

Log likelihood = -100.724

	low	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
age		.9732636	.0354759	-0.74	0.457	.9061578	1.045339
lwt		.9849634	.0068217	-2.19	0.029	.9716834	.9984249
race							
Black		3.534767	1.860737	2.40	0.016	1.259736	9.918406
Other		2.368079	1.039949	1.96	0.050	1.001356	5.600207
smoke		2.517698	1.00916	2.30	0.021	1.147676	5.523162
ptl		1.719161	.5952579	1.56	0.118	.8721455	3.388787
ht		6.249602	4.322408	2.65	0.008	1.611152	24.24199
ui		2.1351	.9808153	1.65	0.099	.8677528	5.2534
_cons		1.586014	1.910496	0.38	0.702	.1496092	16.8134

Note: **_cons** estimates baseline odds.

```

> logistic_model |>
+   tidy(exponentiate = TRUE, conf.int = TRUE)
# A tibble: 4 × 7
  term          estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>   <dbl>   <dbl>
1 (Intercept)    0.186     0.452    -3.71  0.000203 0.0781  0.462
2 ageht          0.988     0.00392   -3.05  0.00225  0.981  0.996
3 sex            1.14      0.106     1.23   0.218    0.927  1.40
4 sbp2            1.00      0.00227     0.404  0.686    0.996  1.01

```